

Introduction to Computer Simulation

Home Work 12

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Question 1

1.a)

From the energy measurements at β_0 an explicit energy histogram $H_{\beta_0}(E)$ was constructed by counting the occurrences of each discrete energy value E . The histogram was reweighted to inverse temperatures $\beta_1 = 0.375$ and $\beta_2 = 0.475$ according to

$$H_\beta(E) = H_{\beta_0}(E) \exp[-(\beta - \beta_0)E],$$

followed by normalization such that $\sum_E H_\beta(E) = N$.

The mean energy and specific heat were then computed from the reweighted histograms as

$$\langle E \rangle_\beta = \frac{\sum_E E H_\beta(E)}{\sum_E H_\beta(E)}, \quad C(\beta) = \frac{\beta^2}{L^2} (\langle E^2 \rangle_\beta - \langle E \rangle_\beta^2).$$

For $\beta_1 = 0.375$ we find a mean energy per spin $\langle e \rangle \approx -1.04$ and a specific heat $C \approx 0.46$, while for $\beta_2 = 0.475$ the mean energy is shifted to lower values and the specific heat increases, approaching the finite-size maximum near β_c . These results are in good agreement with direct simulations and with the exact solution of the two-dimensional Ising model. Deviations are attributable to finite-size effects and the rounded nature of the specific-heat singularity for $L = 16$.

1.b

Using the same simulation data generated at

$$\beta_0 = \beta_c = \frac{1}{2} \ln(1 + \sqrt{2}),$$

the mean energy and specific heat at β_1 and β_2 were determined directly on the fly from the measured energy time series $\{E_i\}_{i=1}^N$ according to

$$\langle e^k \rangle_\beta = \frac{1}{L^2} \frac{\sum_{i=1}^N E_i^k \exp[-(\beta - \beta_0)E_i]}{\sum_{i=1}^N \exp[-(\beta - \beta_0)E_i]}, \quad k = 1, 2.$$

From these moments the specific heat was again obtained via

$$C_\beta = \beta^2 L^2 (\langle e^2 \rangle_\beta - \langle e \rangle_\beta^2).$$

For $\beta = \beta_1 = 0.375$, the on the fly reweighting yields

$$\langle e \rangle = -1.03821, \quad C = 0.46397.$$

These values agree within statistical uncertainty with the results obtained from explicit histogram reweighting in part (a), demonstrating that the on-the-fly method provides an efficient and unbiased estimator for thermodynamic observables close to the simulation temperature β_0 .

1.c

The energy distribution

$$P_\beta(e), \quad e = \frac{E}{L^2},$$

at inverse temperatures β_1 and β_2 can also be obtained directly from the simulation performed at $\beta_0 = \beta_c$. Instead of counting each occurrence of an energy value E_i with weight unity, as done at the simulation point, each measurement is weighted by the Boltzmann factor

$$w_i(\beta) = \exp[-(\beta - \beta_0)E_i].$$

The reweighted energy distribution is thus given by

$$P_\beta(E) \propto \sum_{i=1}^N \delta_{E,E_i} \exp[-(\beta - \beta_0)E_i],$$

where $E = -2L^2, \dots, 2L^2$. For direct comparison with the explicit histograms of problem 22, the normalization was chosen such that

$$\sum_E P_\beta(E) = N.$$

The resulting energy distributions at β_1 and β_2 , obtained in this on-the-fly manner, coincide within statistical uncertainties with those obtained from explicit histogram reweighting. In particular, the distributions at $\beta_1 < \beta_c$ are shifted towards higher energies, while those at $\beta_2 > \beta_c$ are shifted towards lower energies, as expected. This confirms that counting the energy “hits” with weight $\exp[-(\beta - \beta_0)E_i]$ provides an equivalent and efficient method to reconstruct the full energy distribution away from the simulation temperature.